Jun-Gill Kang

jungillkang@gmail.com — South Korea

Agency For Defense Development (ADD), South Korea, Doing military service (mandatory for South Korean)
Webpage: https://jgkang1210.github.io/
Google Scholar: click

RESEARCH INTERESTS

Solving a highly dynamic control problem with the simplest controller. I believe that the combination of model-based control and learning-based, model-free approach can solve highly nonlinear problems (e.g., locomotion, manipulation, and flight control) with one single controller with minimal hierarchy.

EDUCATION

Pohang University of Science and Technology – POSTECH
Convergence IT Engineering, Mechanical Engineering (Double major), BTech, Feb 2022
GPA:4.09/4.3 (1th/9), Graduate representative (1th / All U.G.)

State University of New York at Stony brook (Jul. 2019 – Dec, 2019) – SUNY
GPA:3.96/4.0

EMPLOYMENT

Researcher
2023.06-2026.06(Expected)
Defense AI Center, Agency For Defense Development (ADD), South Korea
Role : Control of self-driving car in off-road environment. Exploration of multi robot in GPS denied cave.

Researcher
2022.06-2023.04
Mechanical Engineering, Korea Advanced Institute of Science and Technology (KAIST)
Dynamic Robot Control and Design Lab (DRCD) by Prof. Hae-won Park.
Role : Control of KAIST HOUND a versatile quadrupedal robot.

SELECTED PUBLICATIONS

Refereed Journal Article

A highly maneuverable flying squirrel drone with agility-improving foldable wings, IEEE Robotics and Automation Letters (RA-L), 2025
D. Lee*, **J. Kang***, S. Han.

Conference Proceedings

Disentangled Multi-Context Meta-Learning: Unlocking robust and Generalized Task Learning. Conference of Robot Learning (CoRL), 2025
S. Kim*, **J. Kang***, T. Kim, and S. Hong.
A Highly Maneuverable Flying Squirrel Drone with Controllable Foldable Wings. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023
J. Kang*, D. Lee*, and S. Han.

In preperation - currently doing rebuttal

Agile perceptive multi-skill locomotion for quadrupedal robots in the wild. Science Robotics
J. Kang*, J. Park*, T. Song, J. Kim, S. Hong**, and H. Park.**

Conference Workshop

PointNT:Point Navigation Transformer. IEEE/RSJ International Conference on Intelligent Robots and Systems Workshop (IROSW), 2025
J. Kang*, T. Kim*, S. Kim, J. Min, S. Hong, and K. Kwak.
Fast, Perceptive Quadrupedal Locomotion in Complex Terrain. IEEE International Conference on Robotics and Automation Workshop (ICRAW), 2024
J. Kang, J. Park, T. Song, and H. Park.

AWARDS AND HONORS

Mu-Eun-Jae Award
Pohang University of Science and Technology – POSTECH, 2022
Select one student every year among all total graduating students.

Talent Award of Korea
Deputy Prime Minister and Minister of Education award, 2020
For the sprite of challenging and continuous efforts on robotics, won 2020 Korea Talent Award. Select only 100 people every year in South Korea from diverse area (Athlete, Researcher, Talents, CEO, etc.)

GRANTS AND FELLOWSHIPS

Creative IT Engineering Scholarship
2019 - 2021
Korea Government
15K USD + Half year study abroad full funding (Stony brook University, NYC)

Research Officers for National Defense Scholarship
2021 - 2023
Korea Government
15K USD

INVITED TALKS

Graduation congratulatory speech
Pohang University of Science and Technology – POSTECH, 2022
As graduation representative gave speech about our future life after graduation.

CONFERENCE ACTIVITY

Papers Presented

A Highly Maneuverable Flying Squirrel Drone with Controllable Foldable Wings. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023
Oral and poster session.

SERVICE TO PROFESSION

Reviewer
IROS (2024, 2025) CoRL (2025)